

1 DATA INTEGRITY

NPD-Rice-COCO-43

Augmentation families grouped
74,700 / 9,355 / 9,333

MultiCrop-88

Duplicates and conflicts excluded
62,214 / 7,787 / 7,769

Agri-Foundation-145k

Fixed development/test partitions
92,571 / 23,140 / 29,041

2 CONTROLLED TRAINING

Pretrained Swin-S

Shared representation
768 dimensions
30 epochs; validation macro-F1

Flat

Condition
cross-entropy

Dual head

Condition + host
cross-entropy

Secondary penalties

MAE or $\text{KL}(q \parallel p^h)$, $\lambda = 1$

3 LOCKED-TEST EVIDENCE

Paired performance

10 seeds per dataset
Accuracy, macro-F1, ChER

Practical equivalence

± 0.25 points
TOST + Holm adjustment

Hierarchy audit

60 dual-head runs
Host accuracy, HDR, TCI

Confidence audit

Validation-only scaling
ECE, NLL, Brier, AURC

90 completed in-domain training runs; manifests fixed before training